CPIT-240 Database

mid 22.33

Q1: Match

A collection of related data.	Database
DBMS software together with data itself.	Database system
Changing data structures and storage organization without change the DBMS	Program-data-independence
The description of a database.	Database Schema
The actual data stored in a database at a particular moment in time	Database State
Specific things or objects in the mini-world that are represented in the database.	Entity
The prosses of transforming requests and results between levels	Mapping

Q2: MCQs

1. Which one is not an actors on the scene:
    a) Database administrators.
    b) Database designer.
    c) Tool developers.
    d) End users.

2. Used by the DBA and database designer to specify the conceptual schema of a database.
    a) Data Definition Language (DDL).
    b) Data Manipulation Language (DML).
    c) Translate Language (TL).
    d) Not of the above.

3. An entity that is member of subclass inherits:
    a) Some attributes of the entity as a member of the superclass.
    b) Some relationships of the entity as a member of the superclass.
    c) All both attributes and relationships of the entity as a member of the superclass.
    d) Not of the above.

4. What is the unique thing that in weak entity?
    a) Dose not have a key attribute.
    b) Have a key attribute.
    c) None of the above.
    d) All of the above.

5. Which of the following is a key from the suprekey {Ssn,Name,Age}?
    a) Student_ID
    b) Student _Name
    c) {Name, Age}

Q3: Compare

Logical data independence	The capacity to change the conceptual schema without change the external schema.
Physical data independence	The capacity to change the internal schema without change the conceptual schema.

Q4: Fill the blank

1. Database administrators : responsible for authorizing access to the database for coordinating and monitoring.
2. Database designers : responsible to define the content, the structure, the constraints, and functions for transactions.
3. End-users: use the data for queries, reports and some of them update the database.
4. System analysts: understand the user requirements.
5. Operators and maintenance personnel: mange the actual running and maintenance of the database system.

Q5: True/ false

Specialization defining a set subclass of a superclass.	True
Cardinality ratio is the maximum number of relationship instance that an entity can participate.	True
In hierarchy, a subclass can be subclass of more than one superclass.	False
A key is a superkey and vice versa.	False
Weak entity have identifying type.	True
An entity can participate only one relationship.	False
Controlling redundancy is advantages of using DB.	False
Physical models description of data with the data value.	False
DB dose not support of multiple views of the data.	False
SQL*plus is example for stand-alone interface.	True
Centralized DBMS combines everything into single system including-DBMS software, hardware, application programs, and user interface processing software.	True

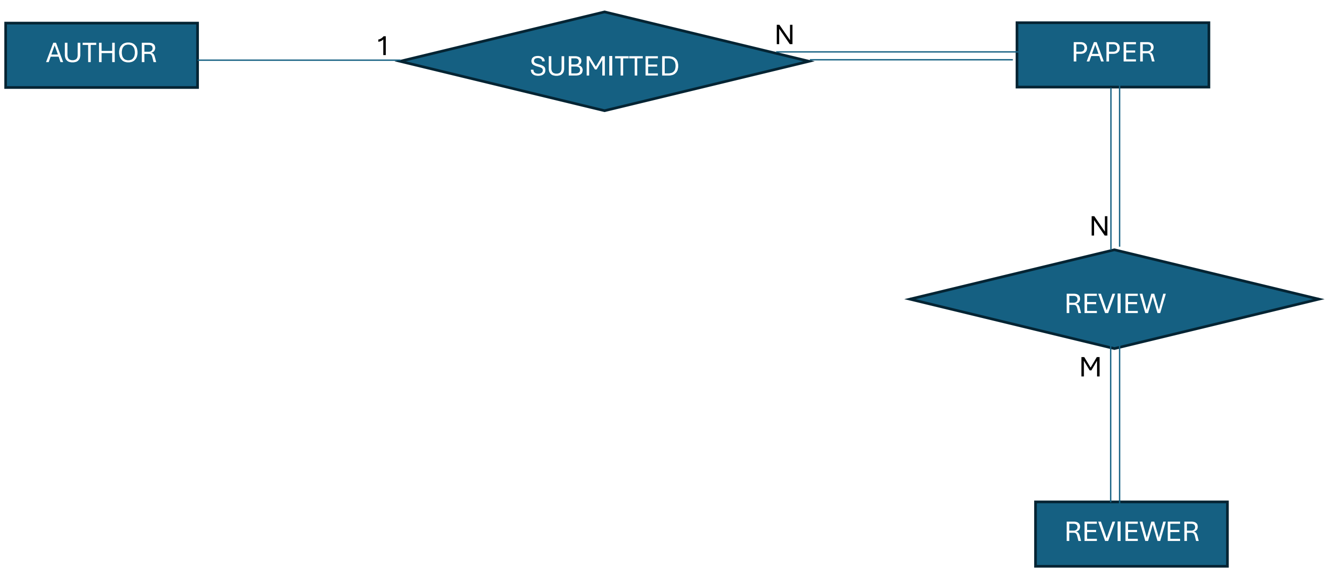
Q6: Draw ER diagram

Consider a CONFERENCE REVIEW database in which researchers submit their research papers for consideration. Reviews by reviewers are recorded for use in the paper selection process. The database system caters primarily to reviewers who record answers to evaluation questions for each paper they review and make recommendations regarding whether to accept or reject the paper. Draw an ER diagram with (participation constraint, Cardinality ratio).

ملاحظة: (السؤال محطوط بشكل تقريبي للي جا فمو كل الشروط محطوطة بس العموم جا شيء بسيط وواضح باينة العلاقات وكل شي حتى المتغيرات مكتوبة باسمها بالصريح)

The data requirements are summarized as follows:

- AUTHOR of papers are uniquely identified by e-mail id. First and last names are also recorded.
- Each PAPER is assigned a unique identifier by the system and is described by a title, abstract, and the name of the electronic file containing the paper.
- REVIEWERS of papers are uniquely identified by e-mail address. Each re-viewer's first name, last name, phone number, affiliation, and topics of in-terest are also recorded.
- A paper may have multiple authors, but one of the authors submitted it.
- Each paper have reviewers at lest one reviewer.



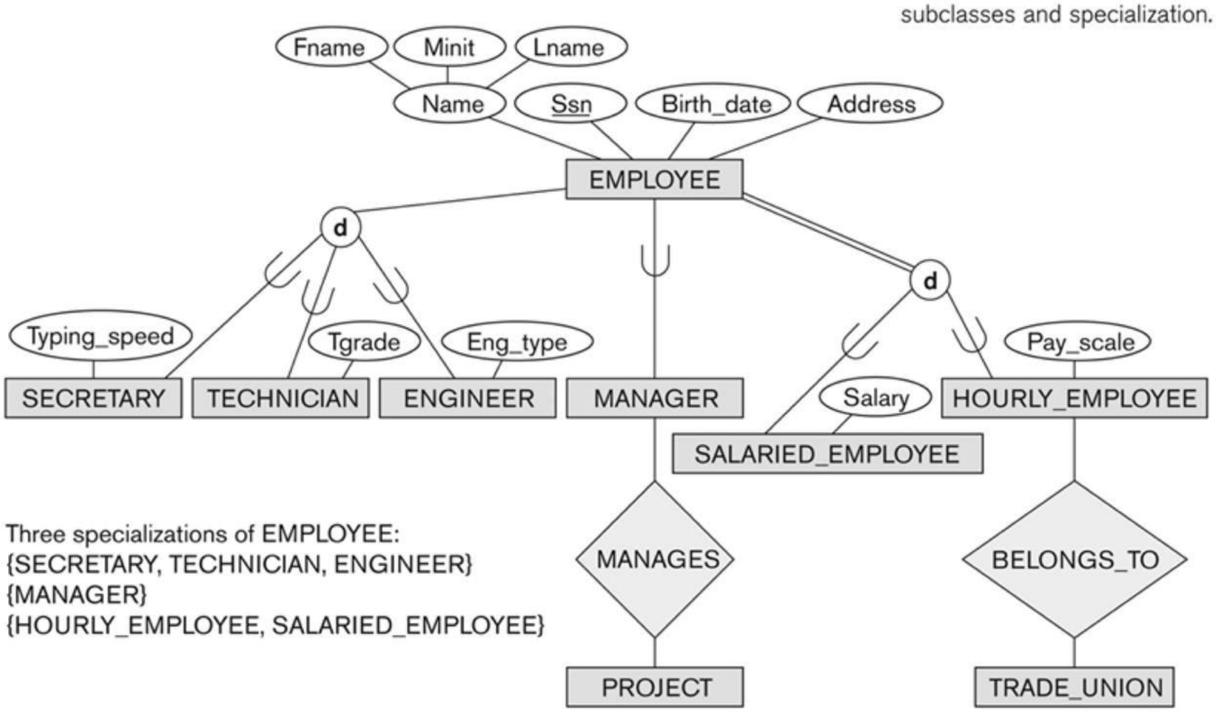
Q7: Complete EER

السؤال عبارة عن صورة لEER Model ، اللي عليك تعبي الفراغ اذا هي disjoint or overlapping

وكمان تحددي الخطوط اذا هي double line or line بحسب الشروط حقت الرسمة

كل شي محطوط وواضح وعليها تكملوا الرسمة ، اليي رح أجطها مو هو اللي جا ولكن تقريبي فتدربوا كثير عليهم.

Figure 4.1
EER diagram notation to represent subclasses and specialization.



Three specializations of EMPLOYEE:
{SECRETARY, TECHNICIAN, ENGINEER}
{MANAGER}
{HOURLY_EMPLOYEE, SALARIED_EMPLOYEE}

Q8: violation

Suppose each of the following update operations is applied directly to the database of the figure, discuss all integrity constraints violated by each operation, if any, and the different ways of enforcing these constraints by answering violation, entity, key, referential, domain and how can solve it.

Insert <'ProductA', 4, 'Bellaire', 2> into PROJECT

violation	Entity integrity	Key constraint	Referential integrity	Domain constraint

Delete <'ProductA', 4, 'Bellaire', 2> into PROJECT

violation	Entity integrity	Key constraint	Referential integrity	Domain constraint

Figure 5.7
Referential integrity constraints displayed on the COMPANY relational database schema.

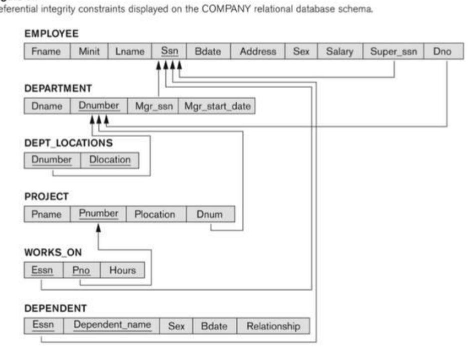


Figure 5.6
One possible database state for the COMPANY relational database schema.

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

بالتوفيق ج~